W4A8 / W8A8 on Ascend

Four bits where the weights are, eight where the tokens are

Zhipu AI · GLM-5 on Huawei Ascend

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Low-Precision & Quantization · arxiv.org/abs/2602.15763

GLM-5 must serve 744B weights from one node

- **744B** params (**40B** active), **256** experts, 75 MoE + 3 dense layers
- Deployment target: one Atlas 800 A3 node, **128 GB × 8**
- Contexts up to **200K** tokens — the KV cache competes with the weights

1.49 TB of weights, 1,024 GB of node

1.49 TB

BF16 weights for 744B parameters, against a 1,024 GB node that also holds the KV cache

Derived: 744e9 × 2 bytes (GLM-5 report Table 10); node = "128GB × 8" from the vLLM-Ascend GLM-5 guide

Both obvious precision plans fail

- **INT8 everywhere** (W8A8)
- Weights **744 GB**: they fit
- **Only ~280 GB left** for the KV cache

- **4 bits everywhere**
- Activations carry **outlier channels**
- One outlier eats the whole INT4 range

Quantize the memory hogs hard, the shared path gently — but 4-bit needs outliers gone first

Insight: the outliers live in the basis

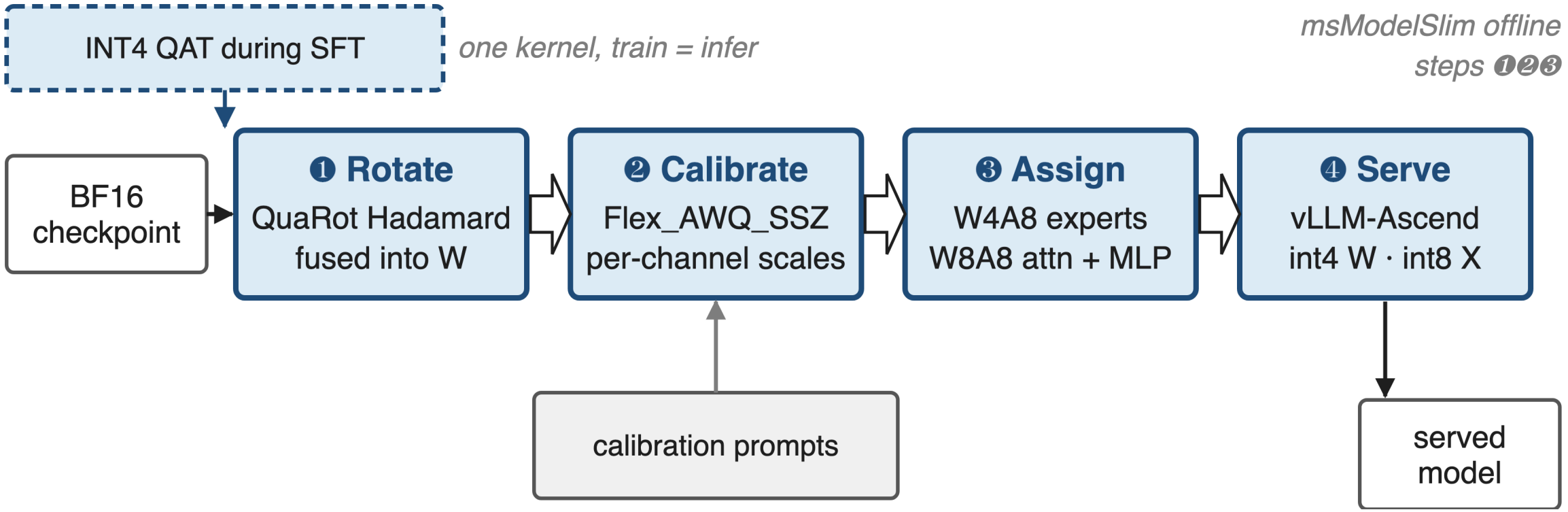
- Outlier channels are a property of the coordinates, not of the information
- An orthogonal matrix's transpose is its inverse, so a rotation changes nothing

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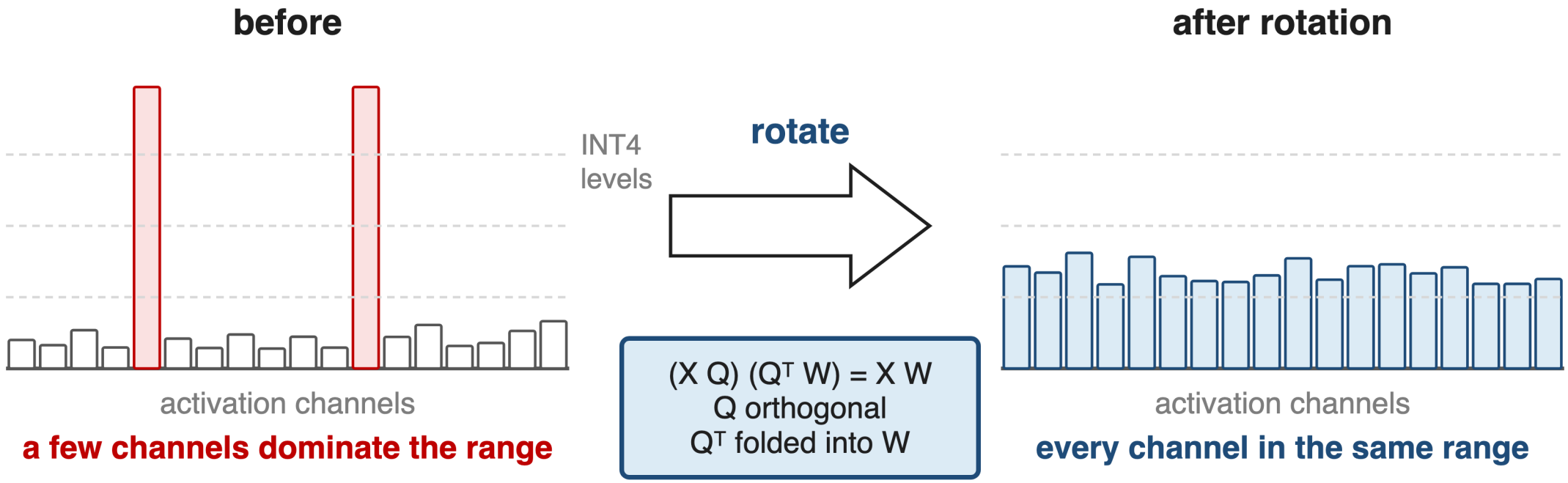
Rotate the model, fold the transpose into the weights, and the outliers are gone for free

Four steps turn BF16 into a served W4A8 model



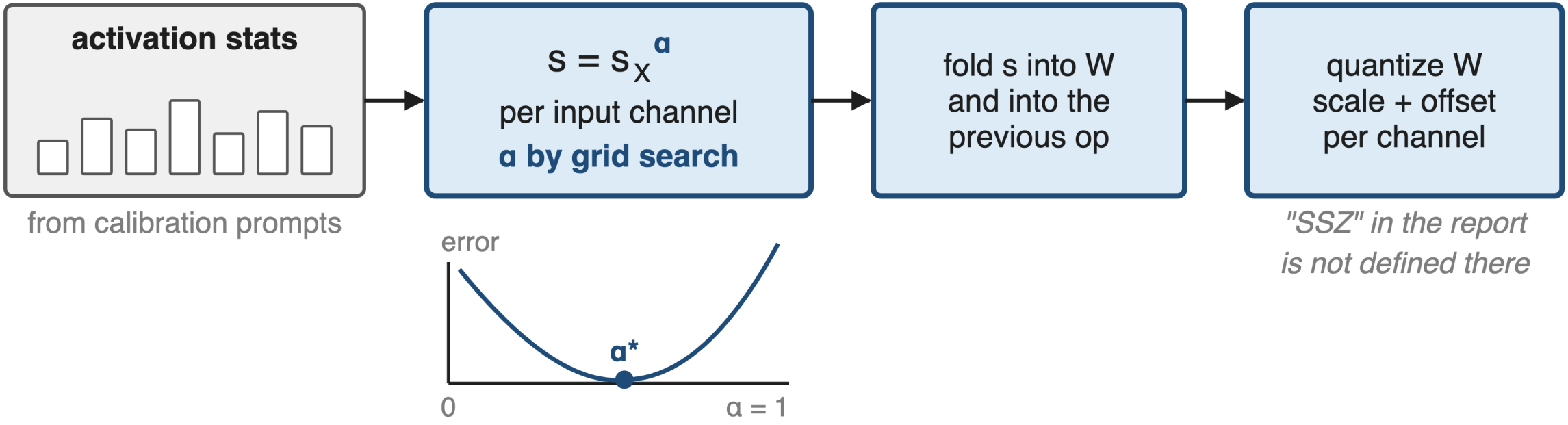
Steps ❶–❸ run offline in msModelSlim (GLM-5 report §5); INT4 QAT is §2.4.3

❶ Rotating the model deletes the outliers



Bar heights are schematic: the point is that no channel keeps the range to itself

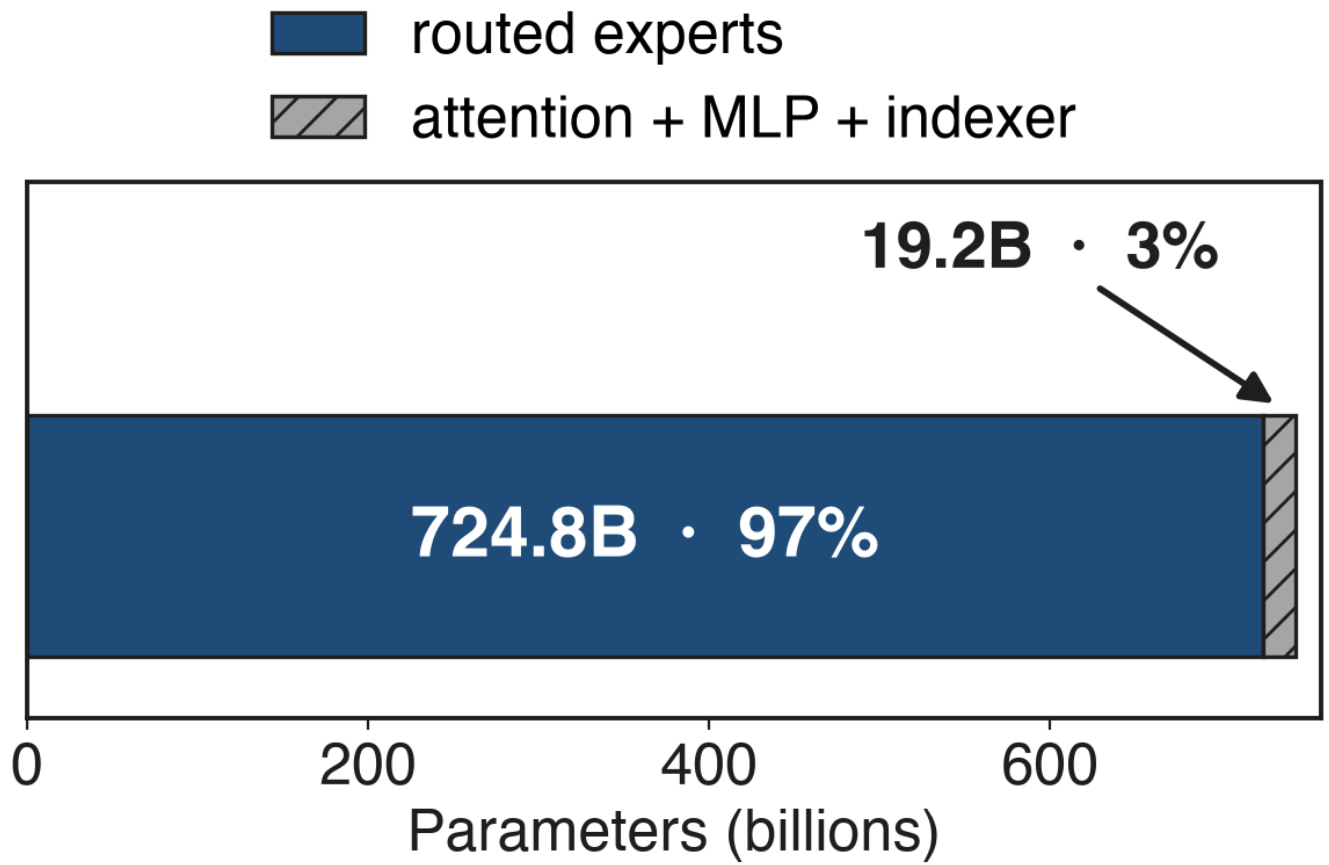
❷ Calibration searches the scales per layer



The report names Flex_AWQ_SSZ but does not define it; the search shown is AWQ's (arXiv 2306.00978, Eq. 5)

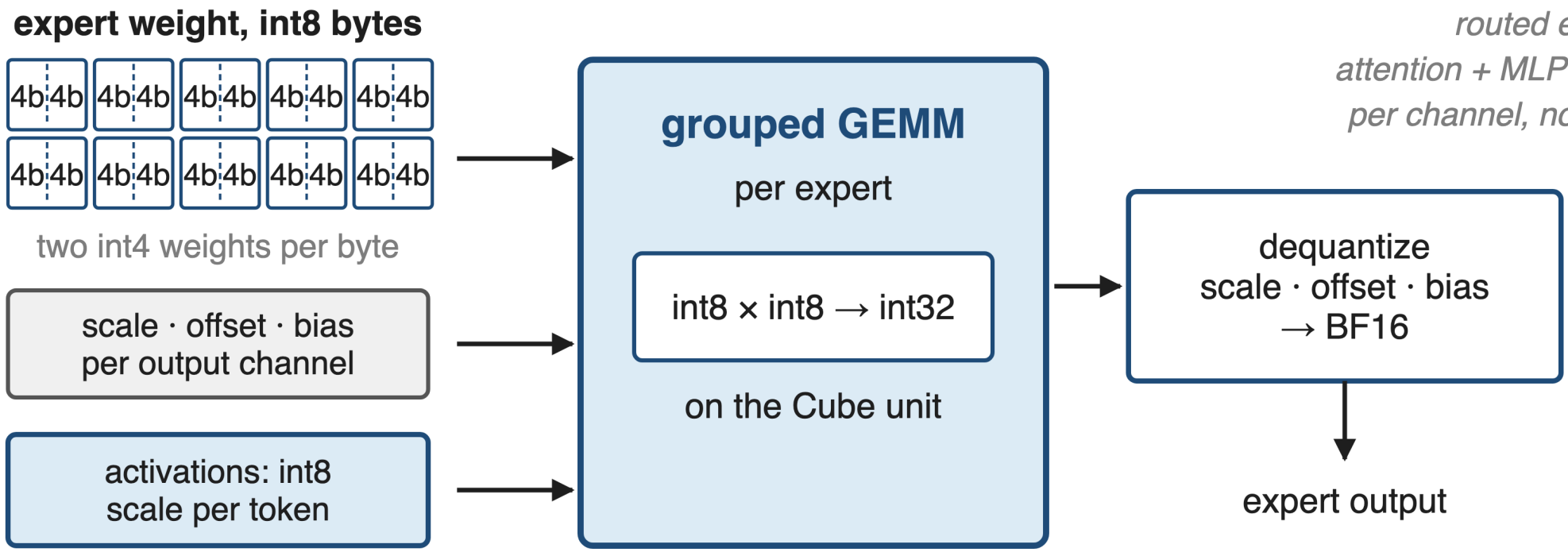
❸ The experts hold 97% of the weights

- Experts: **724.8B** params, few tokens each
- Attention + MLP: **19.2B**, every token
- So: **W4A8** experts, **W8A8** everywhere else



Derived from the GLM-5 report Table 10 — 256 × 3 × 6144 × 2048 × 75 layers

W4A8: int4 weights, int8 activations



The precision plan ships as a per-tensor table inside the checkpoint

❹ INT4 QAT makes training match inference

- INT4 **quantization-aware training** during the SFT stage
- **One kernel** for training and for offline weight quantization

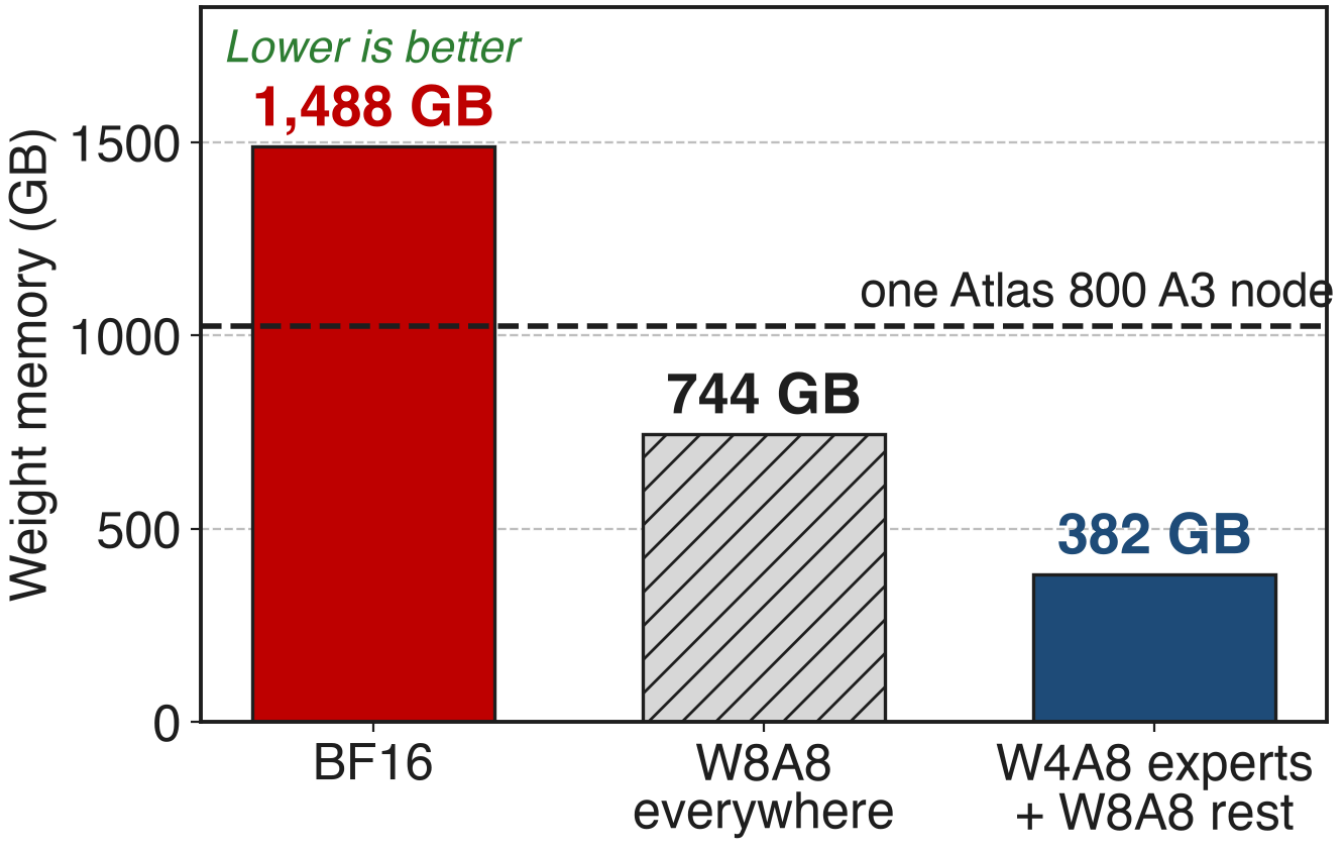
❹ INT4 QAT makes training match inference

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Bitwise-identical behaviour between training and inference, by construction

382 GB of weights, 642 GB free

- **1.95×** smaller than W8A8
- **642 GB** free, not 280 GB
- The guide: use w4a8 weights at **200K** context



Derived from Table 10 parameter counts; node capacity from the vLLM-Ascend GLM-5 guide

The accuracy numbers are not published

- GLM-5 report: W4A8 "without significant accuracy loss" — **no numbers**
- QuaRot's own paper, LLaMA2-70B at 4 bits: **≤ 0.47** ppl, **99%** of zero-shot
- No public accuracy table for the quantized GLM-5 weights

The cost: calibration, QAT, and a frozen plan

- **Calibration + QAT** cost data and training time
- **Per-channel only**: the runtime rejects per-group
- W4A8 covers experts, not attention
- Fused MLA pre-processing: w8a8 by default, "w4a8 may not benefit"
- Re-quantize after any architecture change

Summary

- **1.49 TB** of BF16 weights against a **1,024 GB** node (derived)
- **Insight**: outliers live in the basis — rotate, and 4-bit weights become safe
- Rotate (QuaRot) → calibrate (Flex_AWQ_SSZ) → **W4A8** experts, **W8A8** attention + MLP → INT4 QAT
- **382 GB** of weights, **1.95×** under W8A8; accuracy numbers are not published

https://arxiv.org/abs/2602.15763